Week 1: Course Introduction & Summaries of Data

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STA 220

Winter 2026

Overview

- Welcome to STA220!
- Overview of the course and the syllabus
- Summaries of Data
  - Numerical and graphical summaries of one variable
  - Relationships between two variables

Overview of the Course

Who am I?

- Instructor: Jessie Yeung
- Assistant Professor in statistics
- I've been teaching at UofT for the last 5 years
- Interested in applied social statistics, and bringing statistics to non-statisticians

What is STA220 about?

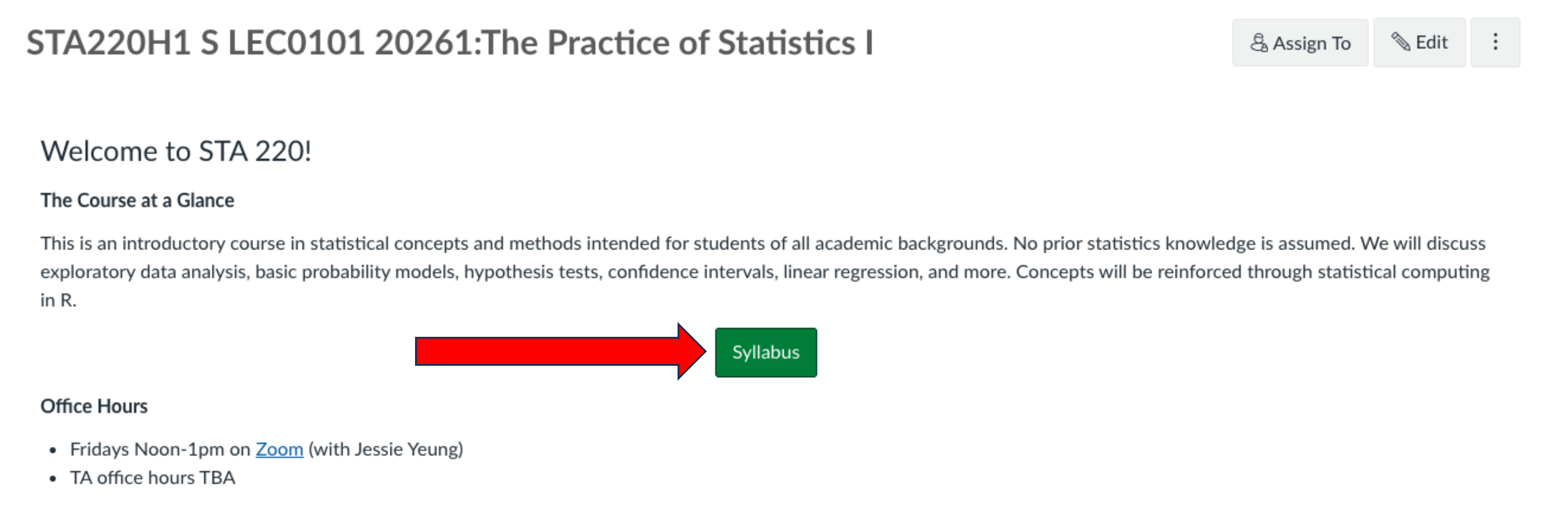
- A first course in statistics
- No prior knowledge of statistics is assumed
- **Statistics**: The process of collecting data, analyzing data, interpreting the results, and presenting conclusions.
- We focus on understanding fundamentals and how to apply statistics (as opposed to mathematical theories)

What is STA220 about?

- In this course, you'll be...
  - Learning the basics of statistical analysis, as well as the reasoning and motivations for them
    - i.e. why do we need statistics?
  - Determining appropriate uses and applications of many common statistical methods
    - i.e. what is the correct tool for my data?
  - Interpreting and using the results of these methods
    - i.e. what do my results mean?
  - Developing statistical literacy, including recognizing the importance of data in decision-making and being a better consumer of information

Syllabus

- https://q.utoronto.ca/courses/426883/files/41380462



STA220H1 S LEC0101 20261:The Practice of Statistics I

Assign To Edit

Welcome to STA 220!

The Course at a Glance

This is an introductory course in statistical concepts and methods intended for students of all academic backgrounds. No prior statistics knowledge is assumed. We will discuss exploratory data analysis, basic probability models, hypothesis tests, confidence intervals, linear regression, and more. Concepts will be reinforced through statistical computing in R.

Syllabus

Office Hours

- Fridays Noon-1pm on Zoom (with Jessie Yeung)
- TA office hours TBA

Communication with the Teaching Team

- All questions about the course content should be posted on Piazza
  - To get there, select ‘Piazza’ from the Quercus menu
  - Questions on Piazza will be answered by the course instructor
- All questions related to course administration should be sent to the instructor at sta220@course.utoronto.ca
- Instructor and TA office hours will be available (see Quercus for schedule)

What is Statistics?

What is data?

- To understand “What is statistics”, we first need to talk about data
- Data refers to a “*collection of numbers, characters, images, or other items that provide information about something*” (DeVeaux et.al. textbook)
- In other words, it is anything meaningful that we can ‘measure’ from an object/person/thing
- This super vague concept of data opens up a lot of questions, like:
  - if data is anything, what is good/meaningful data?
  - once we have data, how do we make any sense of it?

So What is Statistics?

- "*Statistics is the science of learning from data*" - Moore and McCabe, Introduction to Probability and Statistics textbook
- Statistics broadly involves the collection, organization, and interpretation of data, but also much more
- Nearly all fields use statistics: biology, chemistry, medicine, environment, law, politics, etc.
- The goal is the use data to infer facts, make predictions and decisions

Why Statistics?

- The reason for using statistics is that we never get the whole picture.
- The data we have to work with is always some smaller portion of all people/objects/things we could be taking measurements on.
- Most studies want to be able to generalize the results from the data collected to a larger population
  - So obviously the quality of the data impacts the quality of the statistical results.
  - Statistics can therefore help define how to collect good data, so that we get good results.
- But it is always important to be critical!

GFD Example

- 1% of Canadians have celiac disease, and 6% have gluten sensitivities as diagnosed by a physician
- But Gluten-Free Diets (GFDs) were a huge trend, and many people have tried it
- So if you are not gluten-sensitive, is it worth it?
- A U.K. study looked into this: Croall et.al. (2019), "Gluten Does Not Induce Gastrointestinal Symptoms in Healthy Volunteers: A Double-Blind Randomized Placebo Trial ", Gastroenterology
- National Post article on Gluten-Free Diets

GFD Example

- Study wished to determine if GFDs for people without gluten sensitivities has any benefit on their gastro-intestinal (GI) health
- Ran a double-blind randomized trial over a 2 week period
  - half of the 28 subjects used high gluten flour in the 2 week period, while otherwise keeping to their GFD
  - other half used gluten-free flour alongside their normal GFD
- Various indicators of GI health measured before the trial and again at the end
- Result: GI health was statistically the same, regardless of whether there was gluten in the diet or not!

GFD Example

- Here are some questions that might come to mind:
  - How did they pick their subjects?
  - Is 28 subjects enough? How many subjects would we need?
  - What does “statistically the same” mean? What is statistical significance?
  - Which statistical methods are relevant for this data?
  - Does it make sense to generalize these results to Canadians?
- This course will touch on all of these questions, and more!

Summarizing Data: One Variable

height (cm)
160
182
196
172
⋮

Five-Number Summary

Given numeric data, the five-number summary is a very fast and convenient way to summarize the data:

1. **Minimum**: The smallest value
2. **First Quartile (Q1)**: 25th percentile, or the middle number between the minimum and median
3. **Median (Q2)**: The middle/midpoint number
4. **Third Quartile (Q3)**: 75th percentile, or the middle number between the median and maximum
5. **Maximum**: The largest value

Example

Find the 5-number summary for the following data:

1 2 3 4 5 6 7 8 9

10, 10, 11, 12, 13, 14, 14, 14, 19

min = 10

max = 19

median = 13

Q1 = 11

Q3 = 14

Example

What about when the number of data points (n) isn't favourable?

Position = (Percentile / 100) x (n-1) + 1

Find the 5-number summary for the following data:

1 2 3 4 5 6 7 8 9 10

10, 10, 11, 12, 13, 14, 14, 14, 19, 24

median: position $= \left(\frac{50}{100}\right) \times (10-1) + 1$

$= 5.5$

median $= 13.5$

Q1: position $= \left(\frac{25}{100}\right) \times (10-1) + 1$

$= 3.25$

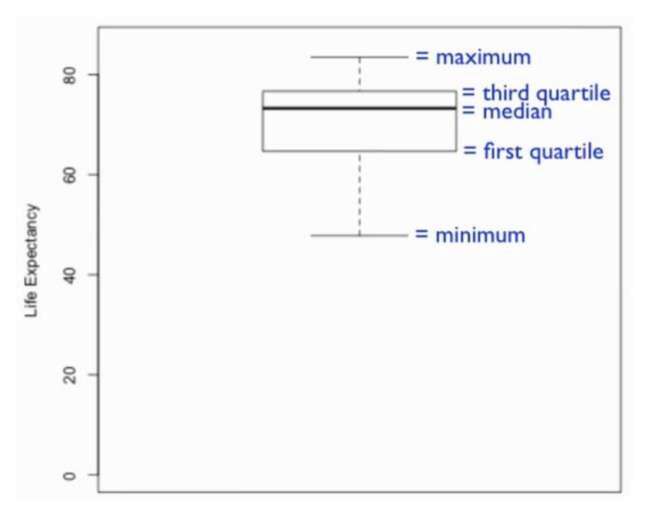
We want the number that is a quarter of the distance from position 3 to position 4.

Q1 = 11.25

- Note: Results may vary when using software to calculate quantiles because there are multiple ways to interpolate between values

Unmodified Boxplots

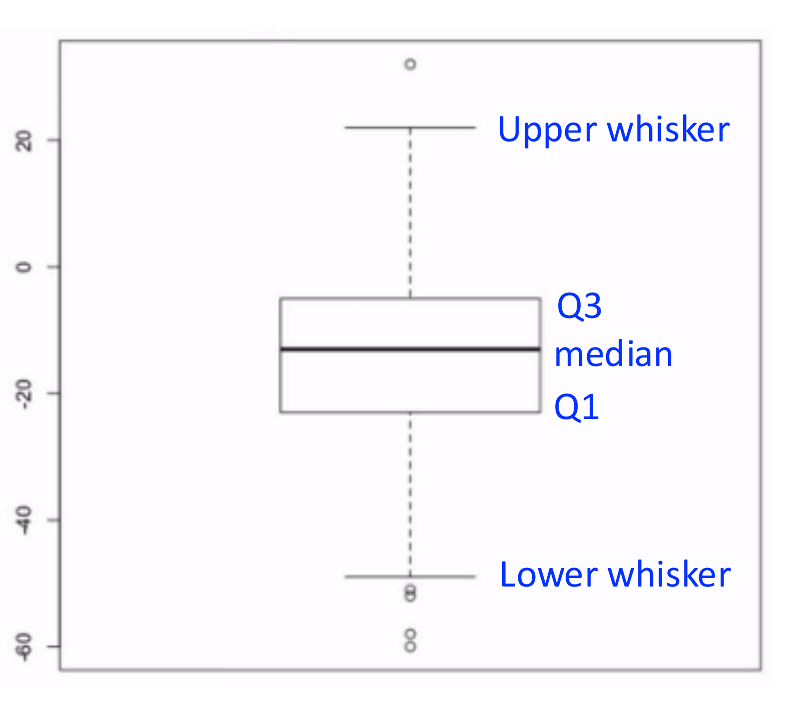
- The five number summary can be visualized using a boxplot (unmodified)



- However, the unmodified boxplot doesn't tell us whether we have extreme values.

Modified Boxplot

- The modified boxplot will show us extreme/unusual values
  - IQR = Q3 – Q1
  - Lower inner fence = Q1 – 1.5 x IQR
  - Upper inner fence = Q3 + 1.5 x IQR
  - The whiskers extend to the largest data value that is less than or equal to the upper inner fence and the smallest data value that is greater than or equal to the lower inner fence.
  - Data values outside the fences are indicated separately as their own points



Modified Boxplot

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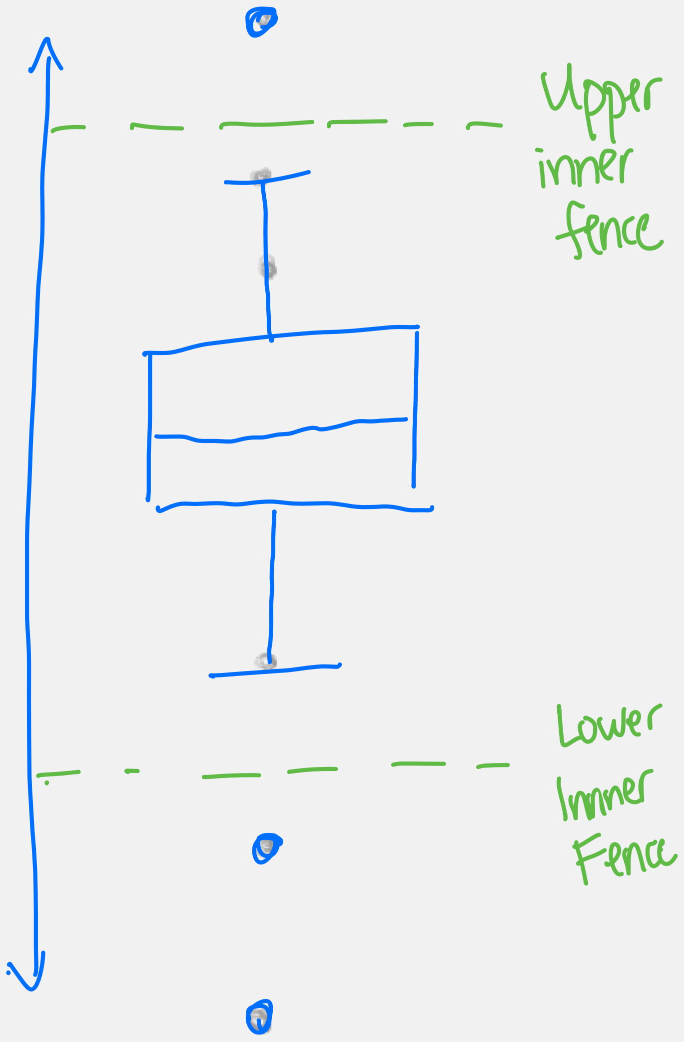


Centre and Spread

- There are two features of our data that we typically want to summarize:
- Where is the centre?
    - Median (50% point), mean (arithmetic average), k% trimmed mean (average of trimmed data)
- How spread out/variable is it?
    - Range (max-min), IQR (Q3-Q1), variance/standard deviation

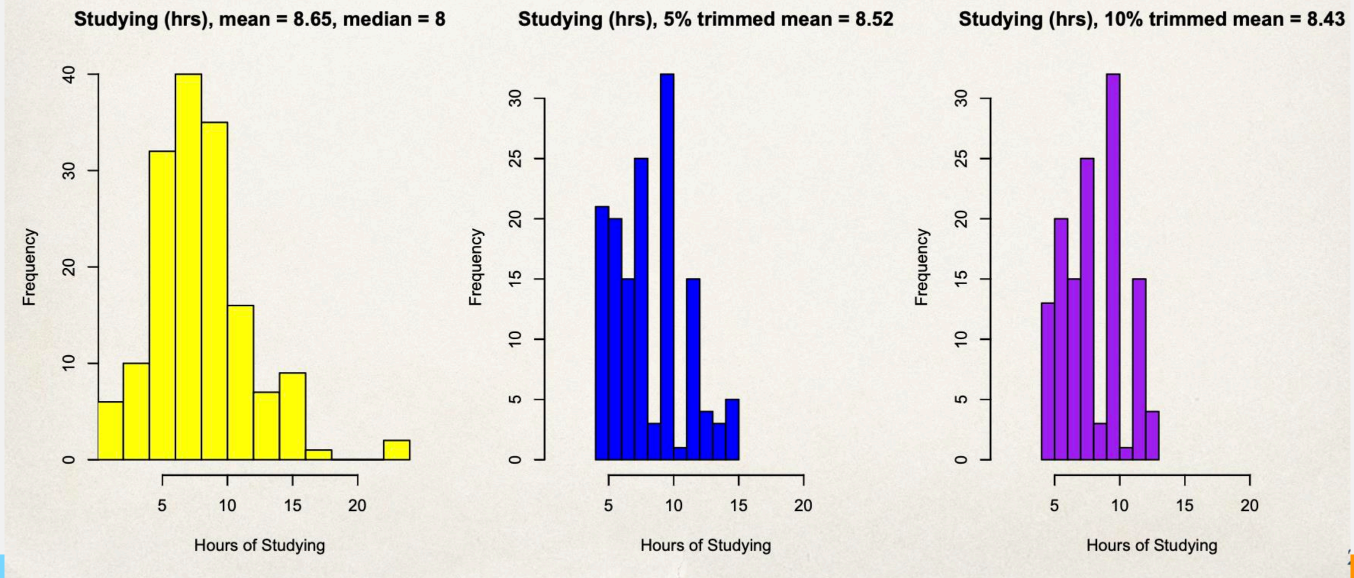
Measures of Centre

$3, 3, 3, 4, 4, 5, 100000000$

- The **median** is the middle value of your data
- The **mean** is just the average of all values of your variable
  - Add up all the values and divide by how many there are
  - Suppose our data values are: $x_1, x_2, \dots, x_n$. The mean is $\bar{x} = \frac{\sum x_i}{n}$ $= \frac{x_1 + x_2 + \dots + x_n}{n}$
  - Limitation: the mean is not a robust statistics because it can be pulled around by very large/small observations
- The **k% trimmed mean** addresses this by removing k% of the largest and k% of the smallest observations
  - If we are trimming the top and bottom of our data by k%, then that means we actually remove 2k% of our data.
  - Then just take the mean of the leftover data.

Measures of Centre: k% trimmed mean

- The more trimming you do, the closer you get to the median.
- Consider a dataset of 158 values
- Here, 5% trimming means removing 0.05x158 = 7.9 so 7 values from each end
- And 10% trimming means removing 0.1x158 = 15.8 so 15 values from each end
- By trimming, you are reducing the variability (spread) of the data
- Consider the following histograms. Notice how the trimmed mean approaches the median

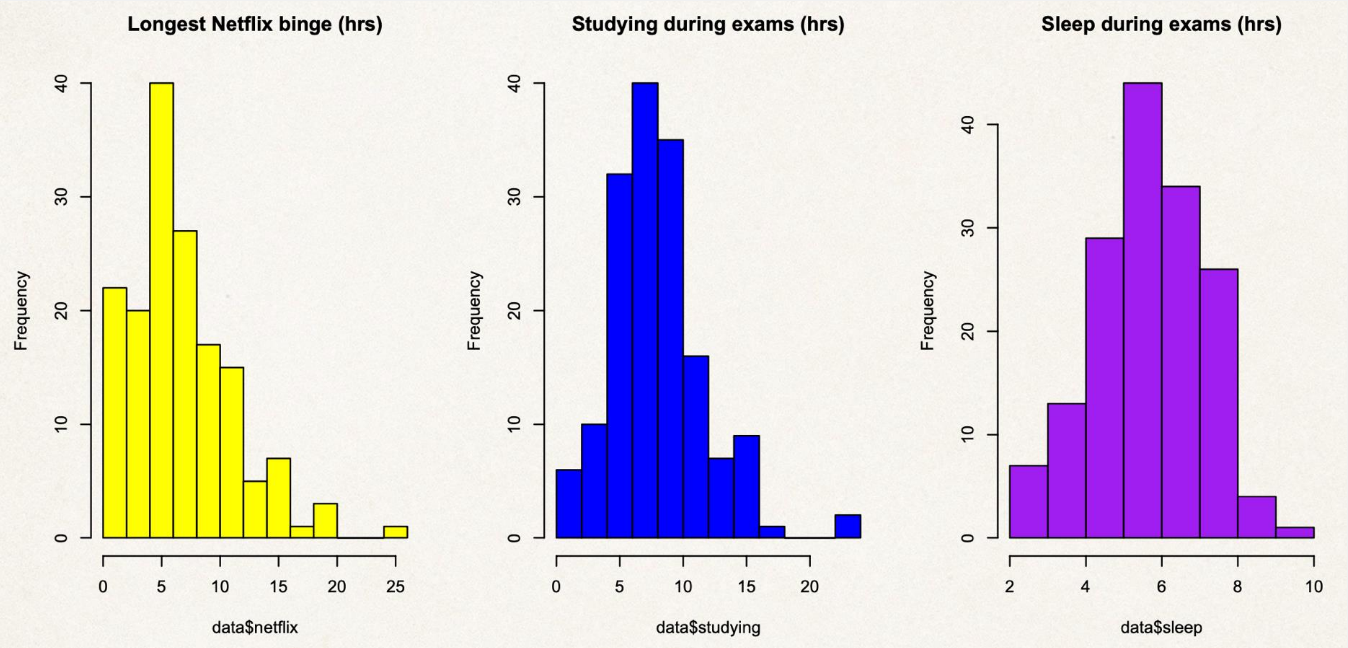


Measures of Spread/Variability

- There are a number of different statistics that we can use to refer to how spread out the data are.
- **Range** of the data: maximum value – minimum value
- **Interquartile Range** (IQR): Q3-Q1
- **Variance** = $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$
- **Standard deviation** = $s = \sqrt{\text{Variance}}$

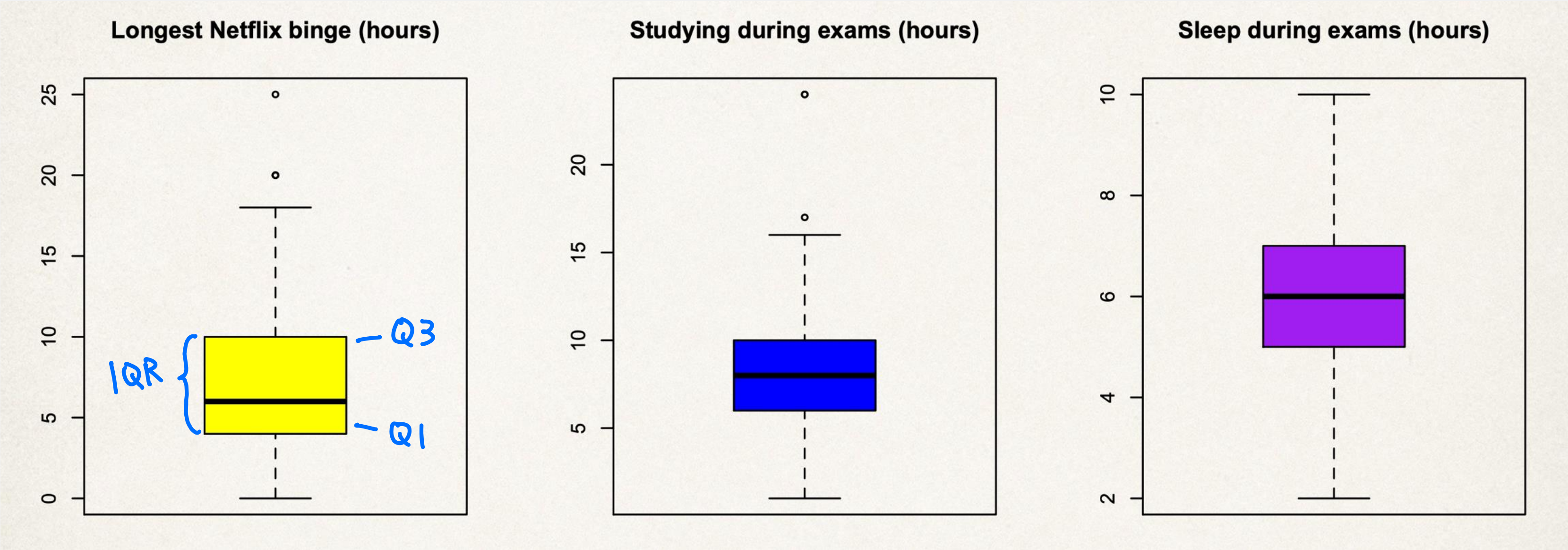
Range of the data

- Range will give you the whole interval where the data sits (usually a very large number)



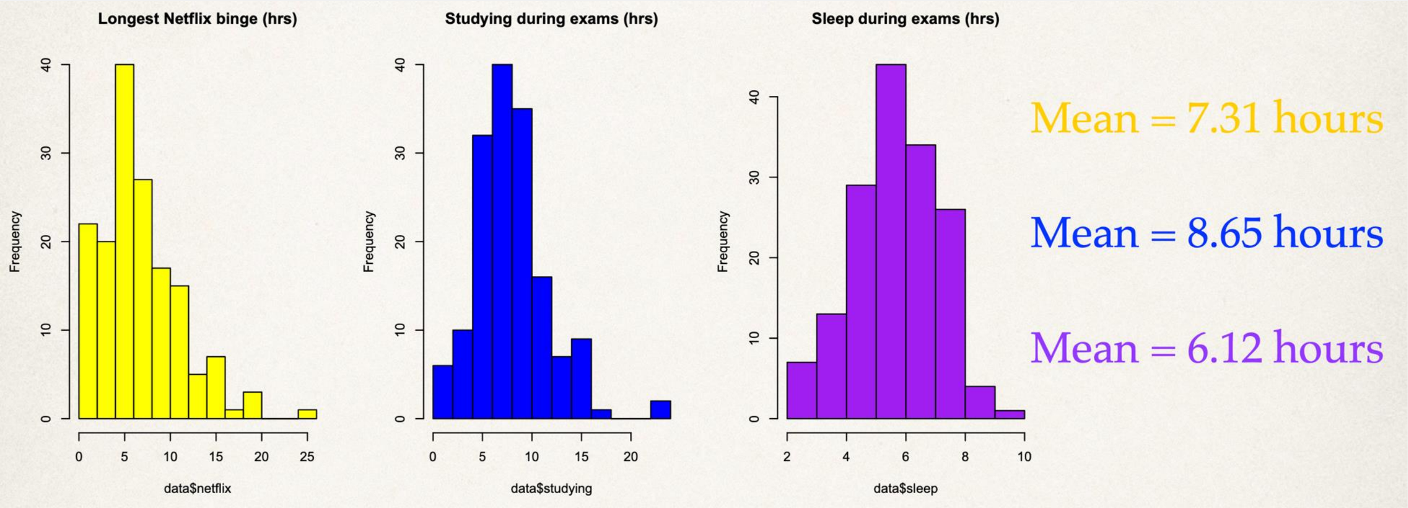
Interquartile Range (IQR)

- IQR gives you the spread of the middle 50% of the data

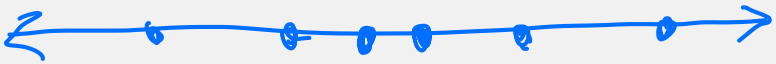


Standard Deviation and Variance

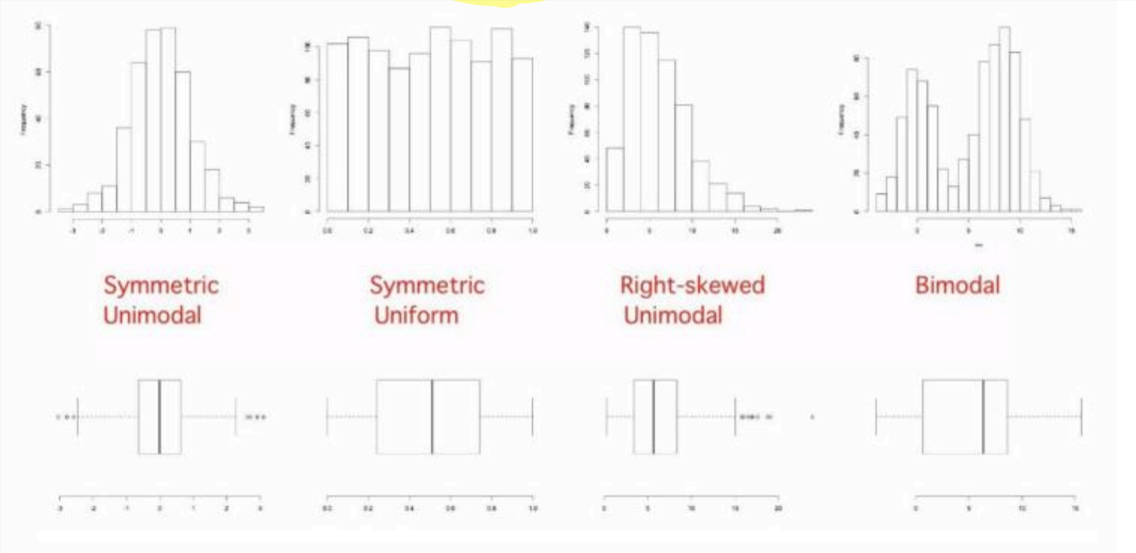
- Variance = $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$
- Standard deviation = $s = \sqrt{\text{Variance}}$
- The standard deviation (roughly) measures the average distance between each point and the mean

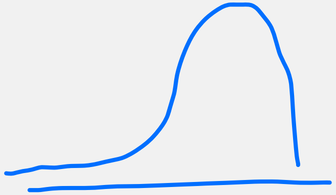


Centre, Spread, and ....



- We can learn a lot about the data by knowing the centre and spread.
- However, there is one more piece that is critical to understanding a dataset's distribution: the **shape** of the data





Left Skewed

- **Distribution**: The pattern of values in the data, showing their frequency of occurrence relative to each other

Z-scores

- Z-scores are a way to compare observations from different datasets
- It is a way of removing the differences between datasets so that we can compare apples to apples
- Example:
  - You received a grade of 80% for a test where the class mean was 70% and a standard deviation was 5%.
  - Your friend in another section with another version of the test also scored 80%. However, your friend's section had a class mean of 76% and a standard deviation of 2%.
  - Who did better?
- You want to know who did better, after taking into the account the performance of each of your sections.

Z-scores

- We can calculate the z-score as follows:

$$\frac{\text{value} - \text{mean}}{\text{SD}}$$

- Calculate the z-score for your grade:

$$\frac{80-70}{5} = 2$$

- Calculate the z-score for your friend's grade:

$$\frac{80-76}{2} = 2$$

- Who did better relative to their test's difficulty?

Both performed equally as well.

Categorical Variables

- We will be talking about two broad categories of data:
  - **Quantitative/numeric**: data that is numeric, generally can be measures with an instrument
    - e.g. age, height, blood pressure, …
  - **Qualitative/categorical**: data that can be grouped into a category/quality
    - e.g. letter grades, true/false, favourite colour …
- How we visualize and summarize data will depend on the type of data we have
- The types of summaries we have seen so far (mean, median, variance, range, histograms, boxplots) will only work for quantitative data
- What types of summaries work for qualitative/categorical variables?

Proportions:

25% said Yes

Bar plot:

Pie Chart:

Distributions of Quantitative and Categorical Variables

- The *distribution* of a variable is the pattern of values in the data for that variable, showing the frequency of the occurrence of the values relative to each other.
  - For quantitative variables, that can be shown by the relative frequencies of all possible values of the variable.
  - For categorical variables, that can be shown by frequencies or relative frequencies of each of the categories of the variable.
- We can also look the distribution of multiple variables jointly

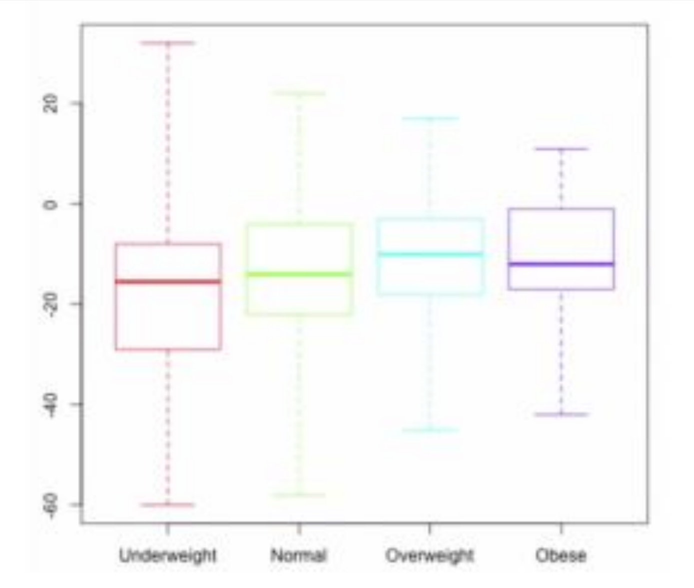
Summarizing Variables: Two Variables

height	weight
160	25
172	33
185	40
⋮	⋮

Relationship between Quantitative and Categorical Variables

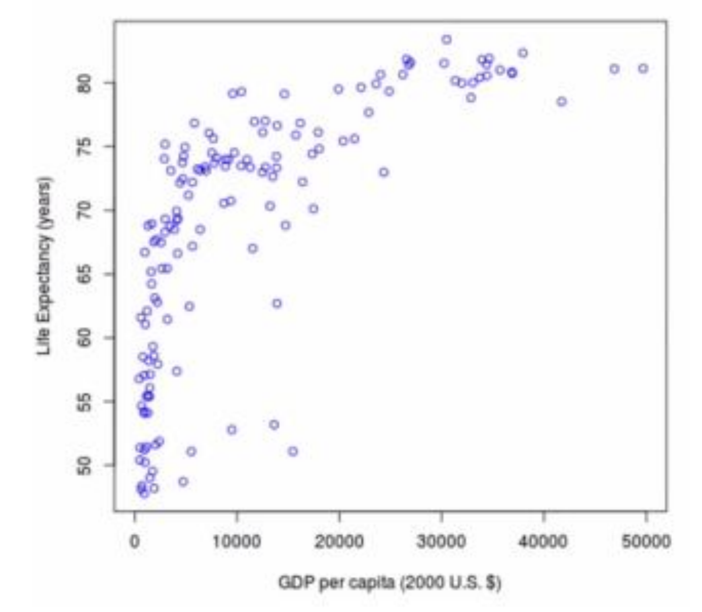
- Compare the 5-number summary or plots for each category

East Asia & Pacific	Sub-Saharan Africa
30 countries and territories	49 countries and territories
median = 72.95	median = 55.44
mean = 73.09	mean = 56.80
min = 62.48	max = 77.65
first quartile = 68.77	third quartile = 59.40



Relationship between Two Quantitative Variables

- Correlation, Scatterplot



Relationship between Two Categorical Variables

- Consider a survey that asks:
  - Do you drink coffee? Yes/No
  - Do you drink tea? Yes/No
- The relationship between two categorical variables can be shown using a **contingency table**

	Coffee - No	Coffee - Yes	Total - Tea
Tea - No	44	29	73
Tea - Yes	40	45	85
Total - Coffee	84	74	158

Relationship between Two Categorical Variables

- We can talk about different types of distributions
- Joint Distribution: this is where we consider both the coffee and tea variables together/jointly (the inside of the table)
- Marginal distribution: we consider only one variable, as if the other one wasn't even there (working only with the outside/margins of the table)

	Coffee - No	Coffee - Yes	Total - Tea
Tea - No	44	29	73
Tea - Yes	40	45	85
Total - Coffee	84	74	158

Relationship between Two Categorical Variables

- Conditional distribution: we condition on one value of one variable (e.g. non-coffee drinkers), and consider what happens with other variable for only these people
  - e.g. out of non-coffee drinkers, how many drink tea?
  - we basically ignore anyone who drinks coffee, as if they were never there
  - so the total number of people we are looking at goes down
  - Out of non-coffee drinkers (84 people), 40 drink tea

	Coffee - No	Coffee - Yes	Total - Tea
Tea - No	44	29	73
Tea - Yes	40	45	85
Total - Coffee	84	74	158

Looking ahead

- Practice Problems and Solutions for this week are posted on the Quercus homepage
- First quiz is due on Sunday, Sept. 11 at 11:59 PM
- Next week we will cover probability